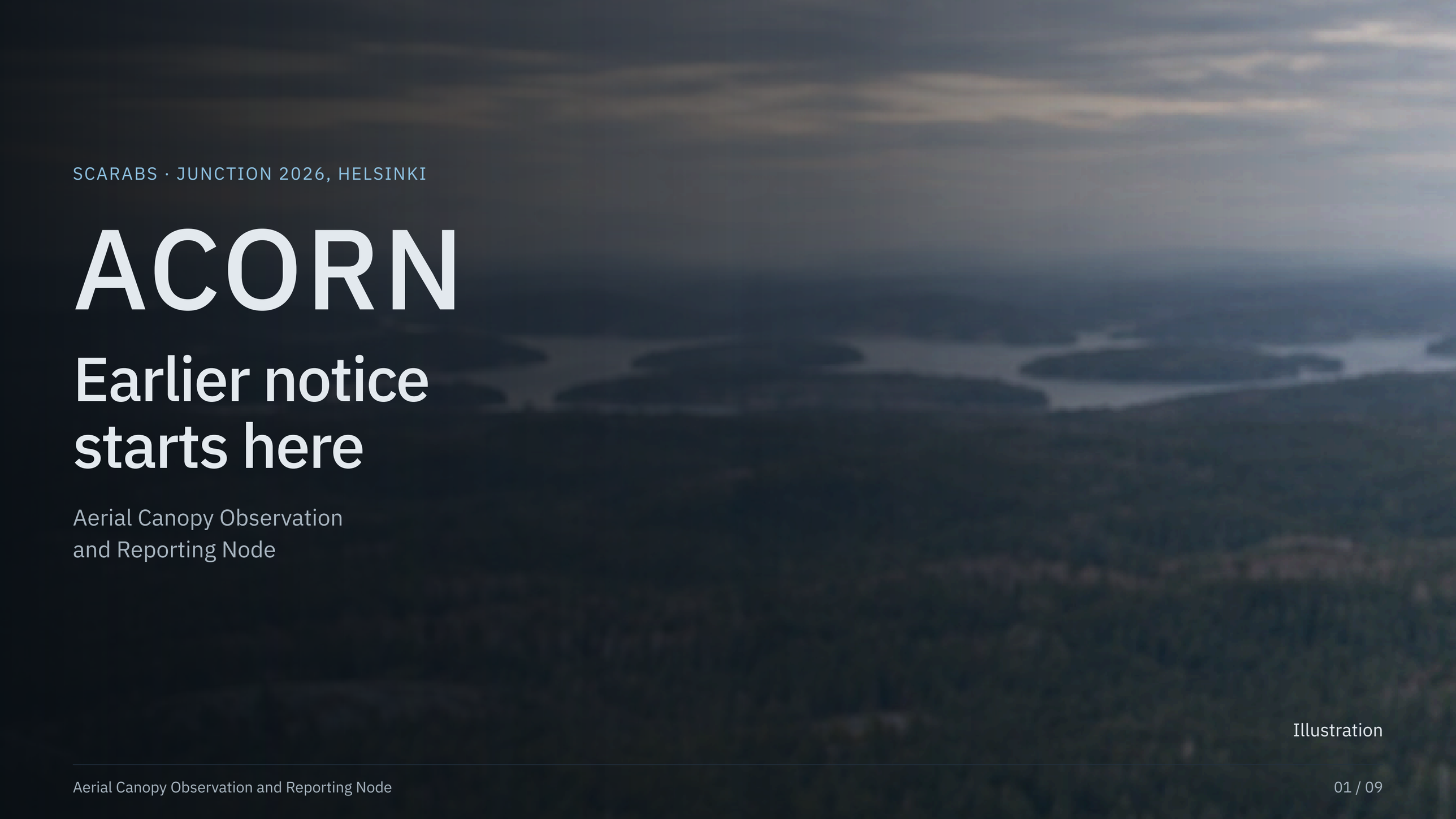
SCARABS · JUNCTION 2026, HELSINKI

ACORN

Earlier notice starts here

Aerial Canopy Observation
and Reporting Node



Illustration

THE QUESTION

Who can keep watch?

Goal: affordable observation without an internet uplink

Cost and field benefit not established



Illustration

A station that keeps analysis local

Two USB cameras

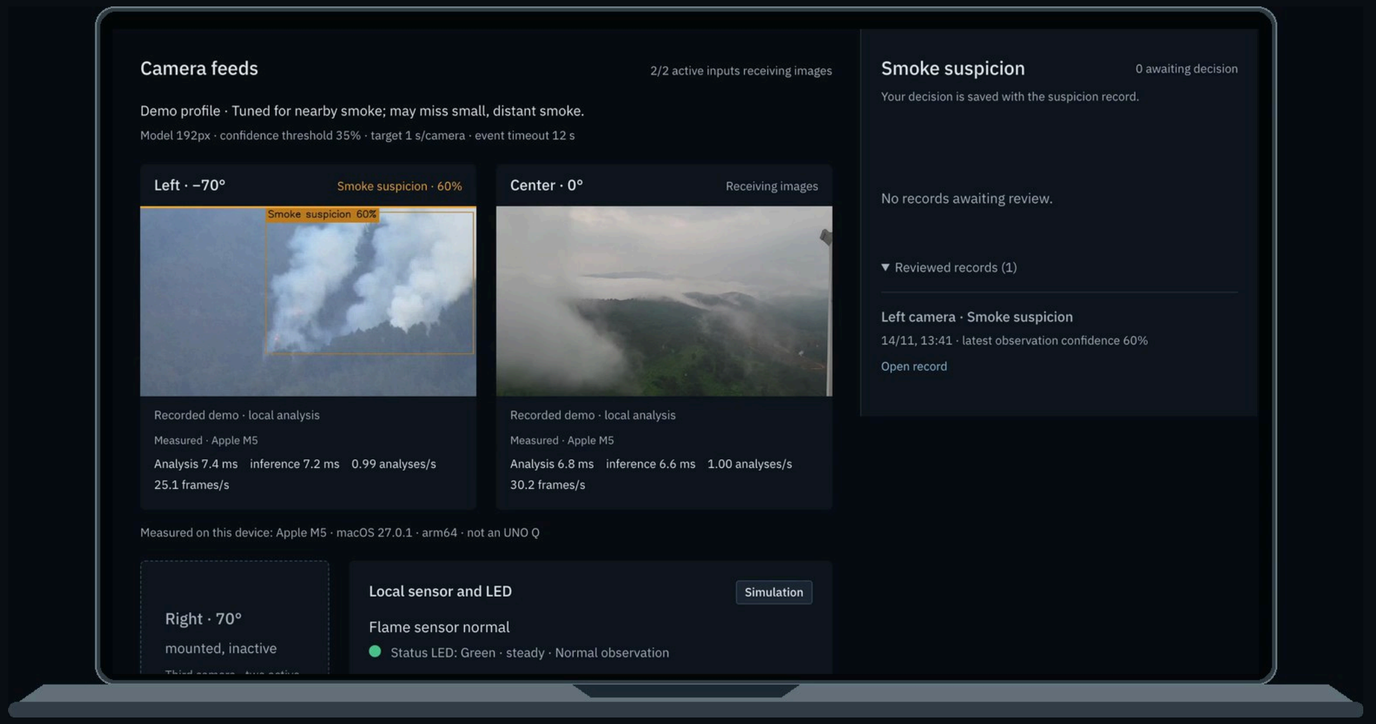
↓

UNO Q Linux

↓

Smoke suspicion

MCU: local IR sensor
+ on-board status LED

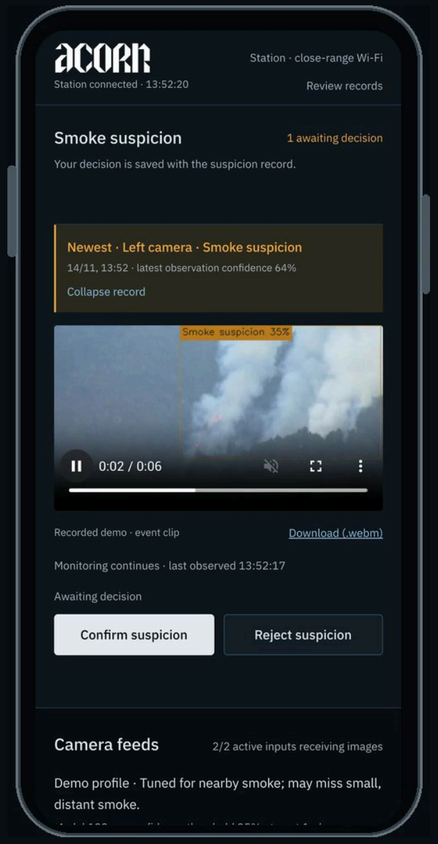


Laptop demo · recorded footage

Documented hardware architecture · assembly photograph still required

A person makes the call

Smoke suspicion → evidence → operator decision



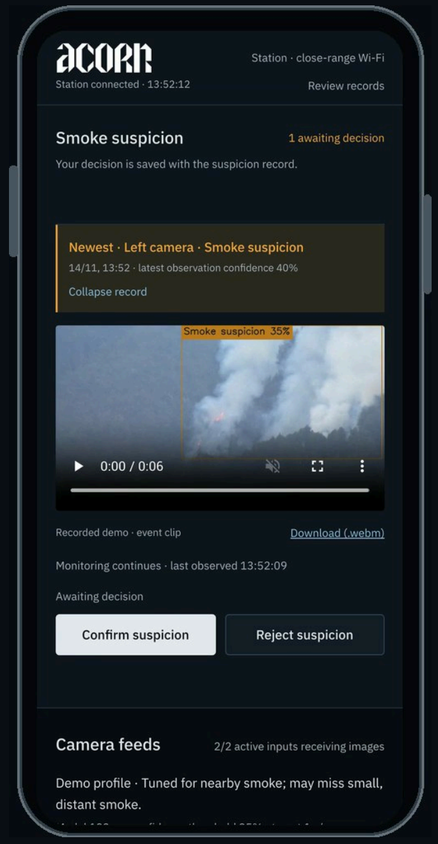
Saved evidence

Smoke suspicion

Confirm suspicion / Reject

Separate states of the same event

Laptop demo · recorded footage



Decision controls

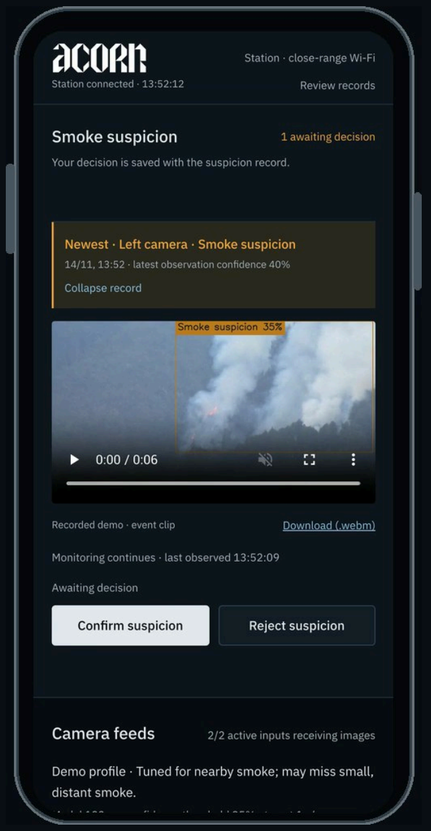
Send the suspicion, keep the imagery

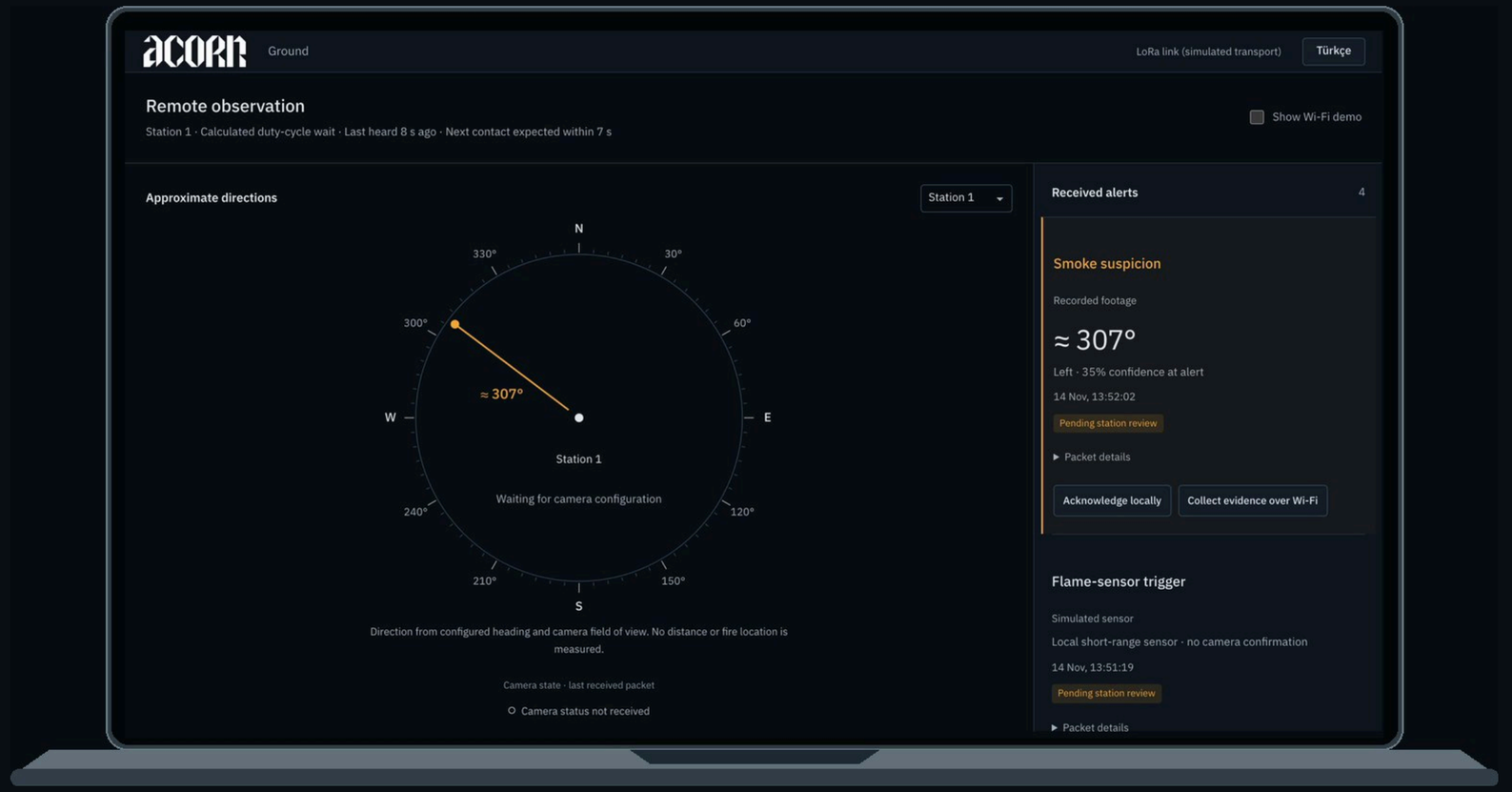
42-byte smoke alert

packet v2 · no imagery over LoRa

LoRa transport
simulated today

Evidence: separate
close-range Wi-Fi



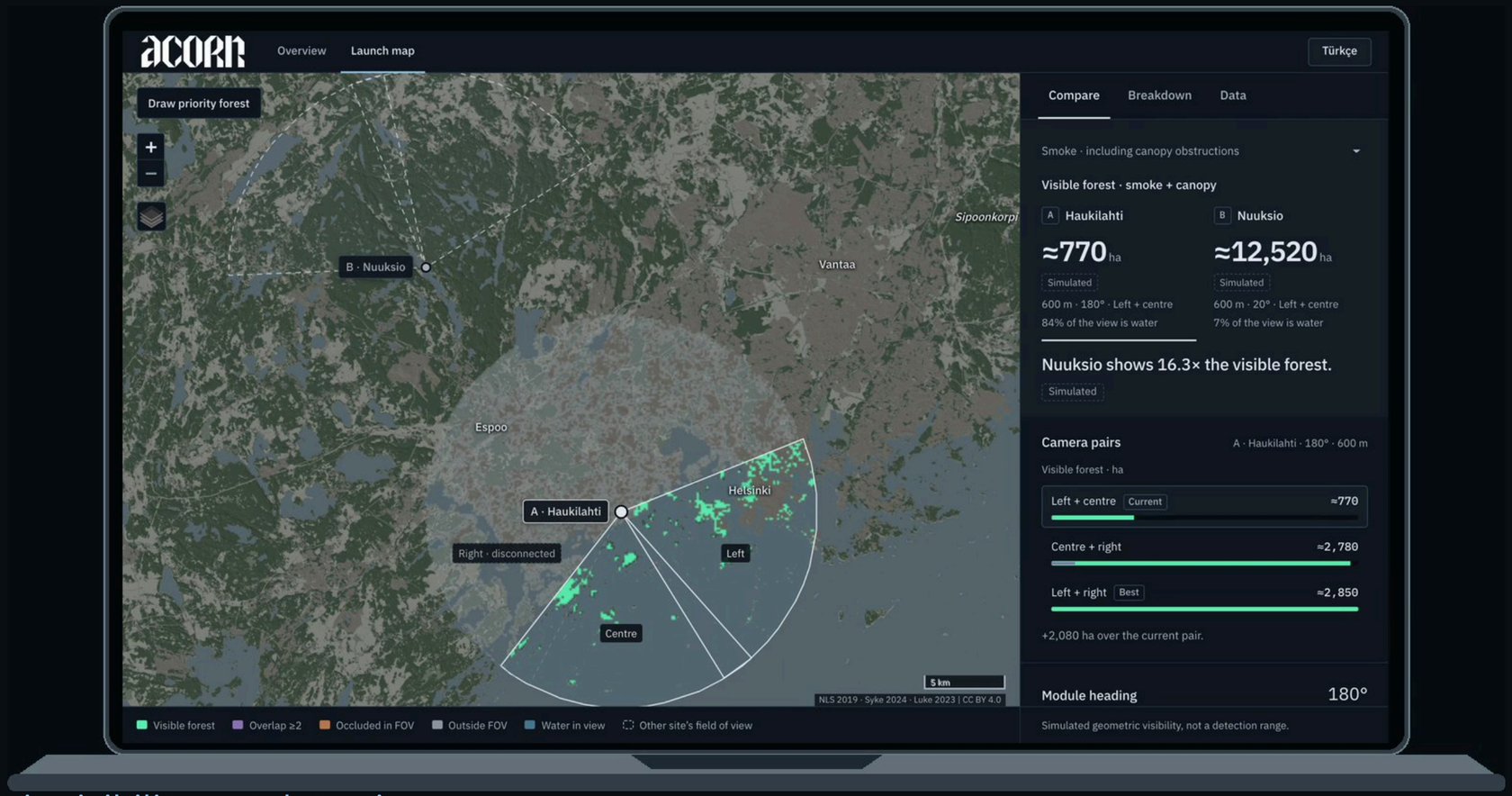


Laptop demo · recorded footage · matching event, Wi-Fi demo off

Put the station where it can see

Real Finnish terrain and forest data

Compare sites · headings · camera pairs



Simulated geometric visibility, not detection range

600 m height · 120 m grid · smoke + canopy · Haukilahti 180° · left+centre · 12 km model limit · 50 m target

Measured, simulated, still open

MEASURED LIVE	**Dense daytime smoke filling the camera view** UNO Q · centre crop · peak scores 0.62–0.76
SIMULATED	**LoRa transport · geometric coverage**
OPEN	**Field sensitivity · real radio · power endurance**

Screen replay · input 192 · threshold 0.35 · Center camera · 3 loops
Scores are not accuracy. Selected news footage failed; no field detection rate established.

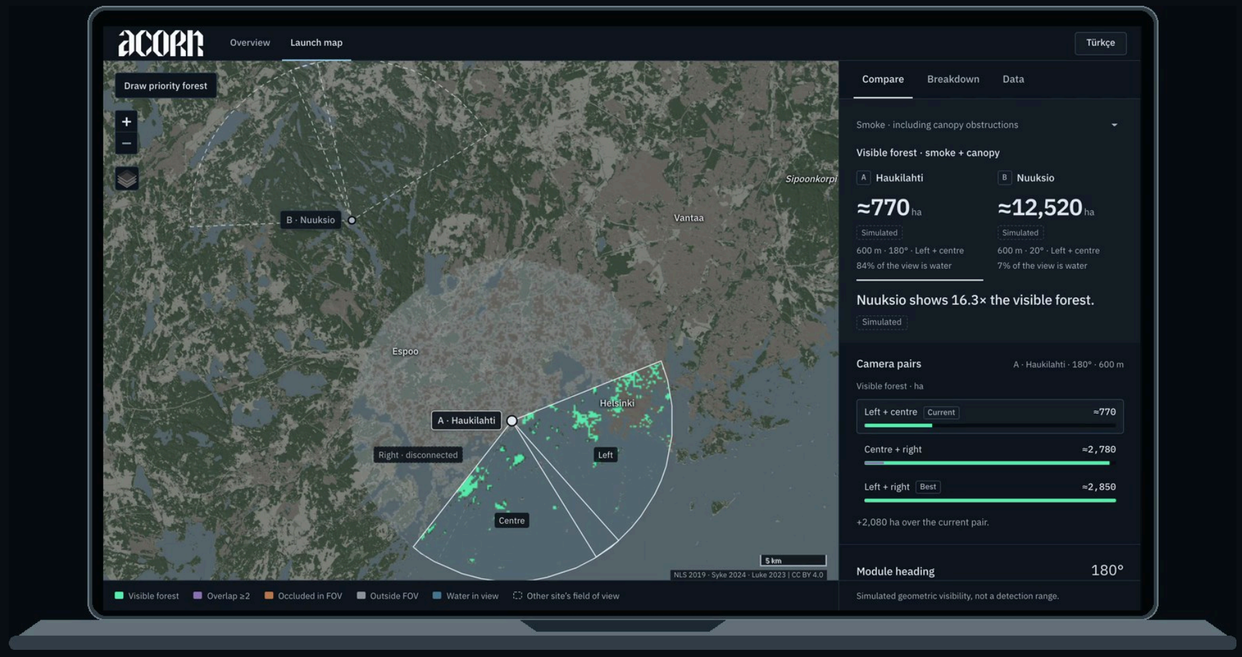
Next proof: a forest trial

Validate board sensitivity

Verify real radio + power endurance

Cost the complete station

Seeking a supervised test setting



Placement simulation · proposed work

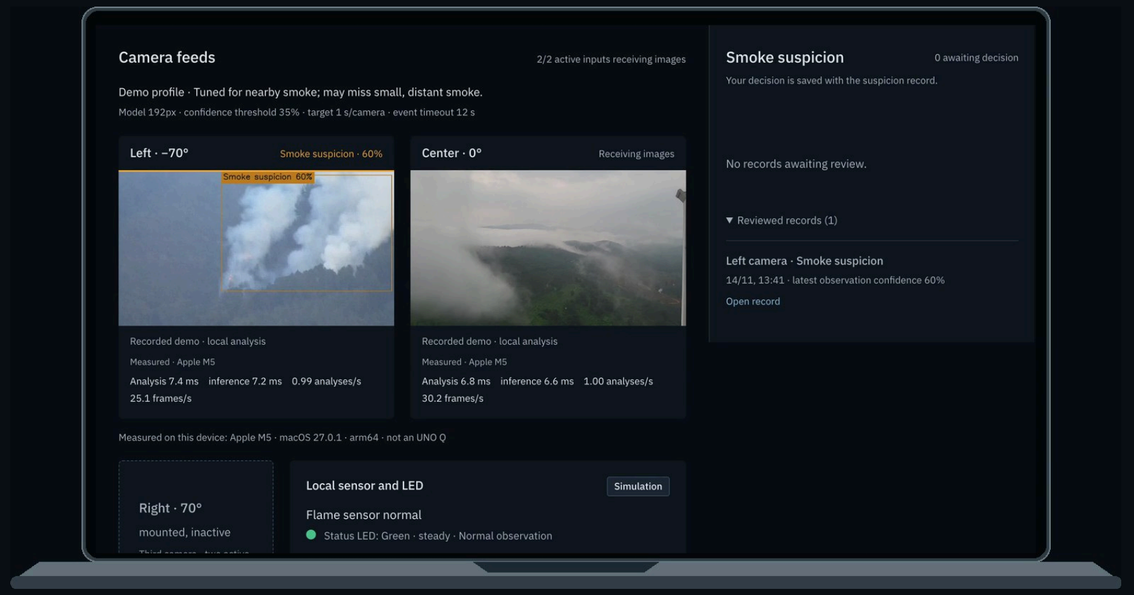
Built for these challenges

Green European Renaissance
forest observation

Best use of Arduino UNO Q
Linux + MCU working together

Best Use of Codex
implementation, tests and board bring-up

Next: live demo



Laptop demo · recorded footage

Scarabs · Junction 2026, Helsinki